

Titanic Dataset – EDA Findings Report

Exploratory Data Analysis | Data Analyst Internship – Task 5

1. Dataset Overview

The Titanic dataset contains 891 passenger records and 12 variables. The analysis was performed using Python, Pandas, NumPy, Matplotlib and Seaborn.

Metric	Finding
Total records	891 passengers
Total variables	12 columns
Survived	342 passengers
Did not survive	549 passengers
Overall survival rate	38.38%
Male passengers	577
Female passengers	314
1st class	216
2nd class	184
3rd class	491

2. Data Quality Findings

- Missing values are present in Age, Cabin and Embarked.
- Cabin has a particularly large amount of missing data, so it requires careful treatment in further analysis.
- Age contains missing observations and should be handled appropriately before modeling.
- Embarked has only a small number of missing observations.
- Duplicate rows were checked as part of the data-quality analysis.

3. Categorical Findings

- The dataset contains more male passengers (577) than female passengers (314).
- Third-class passengers form the largest group with 491 passengers, followed by first class (216) and second class (184).
- Most passengers embarked from Southampton (S = 644), followed by Cherbourg (C = 168) and Queenstown (Q = 77).
- There were 549 non-survivors and 342 survivors.

4. Survival Findings

- The overall survival rate was approximately 38.38%.
- Female passengers had a substantially higher survival rate than male passengers.
- Survival varied strongly by passenger class.
- Approximate survival rates by class were: 1st class 62.96%, 2nd class 47.28%, and 3rd class 24.24%.
- These results indicate that gender and passenger class were important factors associated with survival.

5. Numerical and Visualization Findings

- The Age distribution shows passengers concentrated mainly in younger and middle-age groups, with fewer passengers at very high ages.
- Fare is strongly right-skewed, with most passengers paying lower fares and a smaller number paying substantially higher fares.
- The Fare boxplot shows several high-value potential outliers.
- The Age and Fare boxplots help compare spread, median values and potential outliers across survival groups.
- The Age-versus-Fare scatterplot shows a wide distribution and highlights high-fare observations.
- The correlation heatmap summarizes linear relationships among numerical variables.
- The pairplot provides a multivariate view of distributions and relationships across selected numerical variables.

6. Key Insights

- Gender shows a clear association with survival, with females surviving at a much higher rate than males.
- Passenger class shows a clear survival gradient: first-class passengers had the highest survival rate, while third-class passengers had the lowest.
- Fare is related to passenger class and shows a number of unusually high observations.
- The analysis demonstrates that survival outcomes were not evenly distributed across passenger groups.
- Missing data and outliers should be considered carefully before applying predictive modeling.

7. Conclusion

The Exploratory Data Analysis of the Titanic dataset identified important patterns, relationships and anomalies. The strongest findings concern survival differences by gender and passenger class. The analysis also identified missing values, a right-skewed fare distribution and potential high-fare outliers. Overall, the EDA provides a clear understanding of the dataset and establishes a useful foundation for further statistical analysis or predictive modeling.

Tools: Python • Pandas • NumPy • Matplotlib • Seaborn • Jupyter Notebook